[PHYS 6268] Nonlinear Dynamics: Problem Set 9

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Strogatz Book: Problems 8.1.8, 8.1.11, 8.2.1, 8.2.9, & 8.3.1

1 Problem 8.1.8: Bead on rotating hoop, revisited

The following equation has been derived for the motion of a bead on a rotating hoop:

$$\epsilon \frac{\mathrm{d}^2\phi}{\mathrm{d}\tau^2} = -\frac{\mathrm{d}\phi}{\mathrm{d}\tau} - \sin\phi + \gamma \sin\phi\cos\phi \qquad .$$

Here, $\epsilon \in \mathbb{R}^+$ is proportional to the mass of the bead, and $\gamma \in \mathbb{R}^+$ is related to the spin rate of the hoop. Previously we restricted our attention to the overdamped limit as $\epsilon \longrightarrow 0$.

Now we allow any $\epsilon > 0$.

1.1 (a) Find and classify all bifurcations that occur as ϵ, γ are varied.

We let the typical Newtonian "dot" notation represent differentiation with respect to the nondimensional time coordinate τ, that is, $\frac{\mathrm{d}\phi}{\mathrm{d}\tau} \equiv \dot{\phi}$ for section 1 only. Then, the governing equation for this system can be written as

$$\epsilon\ddot{\phi} + \dot{\phi} + (1 - \gamma\cos\phi)\sin\phi = 0 \qquad .$$

If we let $\vartheta \equiv \dot{\phi}$, then the second order equation of motion can be rewritten as a linear system of two first order equations:

$$\dot{\phi} = \vartheta$$
$$\dot{\vartheta} = -\frac{1}{\epsilon}(\vartheta + (1 - \gamma\cos\phi)\sin\phi)\,.$$

This system admits fixed points (ϕ^*, ϑ^*) at $(n\pi, 0)$ for $n \in \mathbb{Z}$ and along the two-branches curve $(\pm \arccos(1/\gamma), 0)$. γ is restricted to positive values so the solutions for negative γ are not valid. In order to uncover the behavior of the system as the parameters ϵ and γ vary, we analyze the stability in the local vicinity of each fixed point.

To do so, we linearize the system and evaluate the linearized behavior at each of the two fixed points. The linearized system is given by

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\begin{pmatrix}\phi\\ \vartheta\end{pmatrix} = \dot{\boldsymbol{\phi}} = \mathbb{A}\cdot\boldsymbol{\phi} + \mathcal{O}(\boldsymbol{\phi}^2) \quad ; \quad \mathbb{A} = \begin{pmatrix}\partial_\phi\dot{\phi} & \partial_\vartheta\dot{\phi}\\ \partial_\phi\dot{\vartheta} & \partial_\vartheta\dot{\vartheta}\end{pmatrix} = \begin{pmatrix}0 & 1\\ \frac{1}{\epsilon}(\gamma\cos 2\phi - \cos\phi) & -\frac{1}{\epsilon}\end{pmatrix} \qquad .$$

Here, we neglect all higher order terms and their effects on the system's stability. Furthermore, the eigenvalues of the stability matrix $\mathbb{A}$ can be calculated as follows,

$$\lambda_\pm = \frac{\mathrm{Tr}\,\mathbb{A} \pm \sqrt{\mathrm{Tr}\,\mathbb{A}^2 - 4\mathrm{det}\,\mathbb{A}}}{2} \qquad . \tag{1}$$

Since $\arccos(1/\gamma)$ is only defined when $1/\gamma \leq 1$, for this fixed point to even *exist*, the parameter γ must be greater than or equal to unity. The system bifurcates when γ crosses $\gamma_c = 1$, as the topological nature of the system changes with the introduction (or destruction, depending on the "direction") of a fixed point. First, we consider this fixed point which exists for all valid ϵ, but only

$\gamma \geq \gamma_c$. The matrix of stability evaluated at this fixed point reads

$$\mathbb{A}(\arccos\left(\frac{1}{\gamma}\right), 0) = \begin{pmatrix} 0 & 1 \\ \frac{1}{\epsilon}\left(\gamma \cos 2\arccos\left(\frac{1}{\gamma}\right) - \frac{1}{\gamma}\right) & -\frac{1}{\epsilon} \end{pmatrix} \quad .$$

Hence, the trace of the system $\operatorname{Tr}\mathbb{A} = -1/\epsilon < 0$ always, and the determinant is given by

$$\det\mathbb{A} = \frac{1}{\epsilon\gamma} - \frac{\gamma}{\epsilon}\cos\left(2\arccos\left(\frac{1}{\gamma}\right)\right) \quad .$$

This expression for the determinant is equal to zero when $1/\gamma^2 = \cos(2\arccos(1/\gamma))$. This implies that $\det\mathbb{A} = 0$ if

$$\gamma = \sqrt{\frac{1}{\cos\left(2\arccos\left(\frac{1}{\gamma}\right)\right)}} \quad ,$$

which occurs *only* when $\gamma = \gamma_c$. Hence, at the momentary creation or destruction of the fixed point when $\gamma \longrightarrow \gamma_c$, the determinant vanishes and the trace is negative. Therefore, in this case, it is a non-unique fixed point: a stable manifold along the line at $\varphi = 0$. The eigenvalues are given by

$$\lambda_{\pm} = \frac{1}{\epsilon^2} - \frac{1}{2}\sqrt{\frac{1}{\epsilon^2} - \frac{4}{\epsilon\gamma} + \frac{4\gamma}{\epsilon}\cos\left(2\arccos\left(\frac{1}{\gamma}\right)\right)} \quad .$$

The discriminant passes through zero when

$$\epsilon = \frac{\gamma}{4\left(1 - \gamma^2\cos\left(2\arccos\left(\frac{1}{\gamma}\right)\right)\right)} \quad . \tag{2}$$

Otherwise, the determinant is positive, and the the eigenvalues are real and negative. In this case, the fixed points at $\phi^* = \pm\cos\left(2\arccos\left(\frac{1}{\gamma}\right)\right)$ are stable spirals, since again, the trace is always negative. In general, the regions of the phase portrait that are deformed by the stable in-spiraling fixed points grow as γ increases. The notion of a pair of stable fixed points being created and "branching out" from a single, stable fixed point corresponds to the supercritical pitchfork bifurcation.

Now, we turn our attention to the periodic fixed point at the location $\phi^* = n\pi$, such that $\sin\phi^* = 0$. This occurs at the origin as well as multiples of π spreading in positive and negative ϕ. The matrix of stability for this fixed point reads

$$\mathbb{A}(n\pi, 0) = \begin{pmatrix} 0 & 1 \\ \frac{1}{\epsilon}(\gamma \mp 1) & -\frac{1}{\epsilon} \end{pmatrix} \quad ,$$

where the $\mp$ branches correspond to when n is even and odd, respectively. Thus, we recover the same value of the trace $\operatorname{Tr}\mathbb{A} = -1/\epsilon$, so any nodes or spirals at these fixed points are always stable, regardless the parameter values or value of n. The determinant is given by

$$\det\mathbb{A} = \frac{1}{\epsilon}(\gamma \pm 1) \quad ,$$

again where the top branch corresponds to when n is even. Therefore, the topology of this fixed point *also* bifurcates as γ crosses γ_c. When $\gamma < \gamma_c$, the determinant is positive when n is even and negative when n is odd. The eigenvalues are given by

$$\lambda_{\pm} = -\frac{1}{2\epsilon} \pm \frac{1}{2}\sqrt{\frac{1}{\epsilon^2} - \frac{4}{\epsilon}(\gamma \pm 1)} \quad ,$$

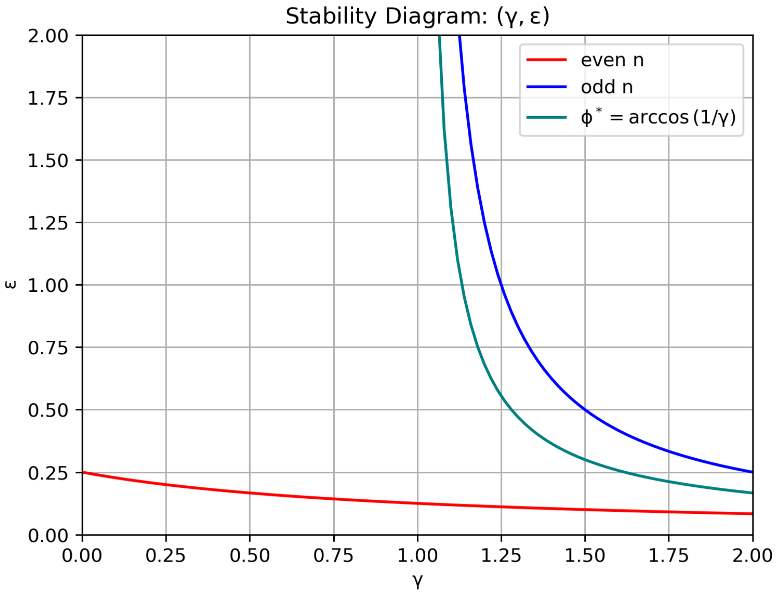

Figure 1: Stability diagram for the system $\epsilon\ddot{\phi} + \dot{\phi} + (1 - \gamma\cos\phi)\sin\phi = 0$, with fixed points at values of $\dot{\phi} = 0$ and $\phi^* = n\pi$ or $\phi^* = \pm\arccos(1/\gamma)$. The red and blue curves correspond to the manifold distinguishing stable and unstable regions for the even and odd values of n, respectively. On the other hand, the teal curve represents the manifold separating stable spirals and unstable saddle points for the alternative fixed point par. Please note that the pair of fixed points is only defined when $\gamma \geq 1$.

which implies that our stability changes when the discriminant vanishes, or when the eigenvalues change from real to complex. This occurs at

$$\epsilon = \frac{1}{4(\gamma \pm 1)} \qquad , \tag{3}$$

or inverted, when

$$\gamma = \frac{1}{4\epsilon} \mp 1 \qquad .$$

Therefore, our system exhibits either stable in-spirals at these periodic fixed points, or saddle nodes. This can be seen by considering the following (fix $\epsilon = 1$): as the parameter γ is increases from, e.g., slightly larger than unity to $5/4 = \frac{1}{4\epsilon} + 1$ the fixed point at $\phi = 0$ changes from a stable spiral to a saddle point. That is, the eigenvalues change from complex conjugates with a negative real part to a pair of real values with the opposite sign. This means that the flow in phase space transitions from a stable inspiraling behavior (like fluid flowing down the drain) to a flow over a saddle valley. Continuing the sink bowl analogy, this is akin to extruding the vertical height of the sink bowl surface along the dimension normal to the faucet extension, turning the flow around the drain from stable to unstable. This evolution near the origin is consistent with the supercritical pitchfork bifurcation.

1.2 (b) Plot the stability diagram in the positive quadrant of the ϵ, γ plane.

The stability diagram is given in Figure 1. The curve from equation (2) is in teal, and the pair of curves corresponding to even and odd n fixed points from equation (3) are shown in red and blue, respectively.

2 Problem 8.1.11: Isothermal Autocatalytic Reactions

In a study of isothermal autocatalytic reactions, Gray and Scott (1985) considered a hypothetical reaction whose kinetics are given in dimensionless form by

$$\dot{u} = a(1-u) - uv^2 \quad , \qquad \dot{v} = uv^2 - (a+k)v \quad ,$$

where $a, k > 0$ are real-valued parameters.

Show that saddle-node bifurcations occur at $k = -a \pm \frac{\sqrt{a}}{2}$.

We first calculate the fixed points of the system in order to linearize about them and assess their stability. By inspection, fixed points occur only at $(u^*, v^*) = (1, 0)$. The linearized system for a general coordinate (u, v) reads

$$\mathbb{A}(u,v) = \begin{pmatrix} -a - v^2 & -2uv \\ v^2 & 2uv - (a+k) \end{pmatrix} \quad .$$

Evaluating this at our fixed point, we find

$$\mathbb{A}(1,0) = \begin{pmatrix} -a & 0 \\ 0 & -(a+k) \end{pmatrix} \quad ,$$

which admits a trace $\operatorname{Tr} \mathbb{A} = (-a - a - k) = -(2a + k)$ and a determinant $\det \mathbb{A} = a(a + k)$. Our eigenvalues readily follow from these properties. It is worth noting that the value of u in the stability matrix is always coupled to v, i.e., it never appears in an isolated manner. Hence, the stability of the system is *not sensitive* to the precise value of the u coordinate, so long as the v coordinate vanishes.

Rather than taking the algebraic approach here, we consider the nullclines and their intersection to determine their intersection's dependence on the parameters. With the variation of a system parameter (while the other is held constant), the fixed points will eventually coalesce along the intersection of nullclines, provided one's behavior is non-monotonic. Hence, where the fixed points converge and "collide" along the system nullclines, our bifurcation occurs.

The nullclines for the system read

$$\begin{aligned} 0 &= a(1-u) - uv^2 \\ 0 &= uv^2 - (a+k)v \end{aligned}$$

The lower equation implies that

$$uv - (a+k) = 0, \qquad \Longrightarrow \qquad v = \frac{a+k}{u}$$

From the top expression, we find that

$$v^2 = \frac{a}{u}(1-u) \qquad \Longrightarrow \qquad v = \sqrt{\frac{a}{u}(1-u)} \qquad .$$

Now that we have isolated both expressions in terms of the v coordinate, we equate the two to determine a condition on the parameters a and k for the intersection of the nullclines. This yields

$$\begin{aligned} \frac{a+k}{u} &= \sqrt{\frac{a}{u}(1-u)} \\ \implies a+k &= \sqrt{au(1-u)} \\ \implies a+2k+\frac{k^2}{a} &= u(1-u) \\ \implies a+2k+\frac{k^2}{a} &\equiv C = u-u^2 \\ \implies u^2-u+C &= 0\,. \end{aligned}$$

Therefore, the roots of the expression $u^2 - u + C = 0$ provide the locations (in terms of $C(a,k)$) where the nullclines' intersection changes from existing in the reals to the complex plane. Hence, the roots of this equation

$$u^* = -\frac{1}{2} \pm \frac{1}{2}\sqrt{1-4C}$$

identify the locations where the nullclines intersect. The v coordinates can be obtained by substituting u^* into either of the expressions for v. Since we are interested in the location of the *bifurcation*, not only the intersection of the nullclines (the fixed points), we now consider the moment when these roots cease to exist. That is, we are interested in the values of k where the discriminant

$$1-4C = 1-4\left(a+2k+\frac{k^2}{a}\right) = 0 \qquad .$$

This condition yields a polynomial in k

$$4k^2 + (8a)k + (4a^2 - a) = 0 \qquad .$$

This quadratic expression is satisfied for

$$-a \pm \sqrt{\frac{a}{4}}, \qquad \implies \qquad k_c = -a \pm \frac{\sqrt{a}}{2} \qquad .$$

Hence, we have shown that the location where the fixed points are annihilated (or created) by the intersection of the nullclines occurs at $k_c = -a \pm \frac{\sqrt{a}}{2}$.

3 Problem 8.2.1: Hopf Bifurcations

Consider the biased van der Pol oscillator

$$\ddot{x} + \mu(x^2-1)\dot{x} + x = a \qquad .$$

Find the curves in (μ, a) space at which Hopf bifurcations occur.

To isolate and identify Hopf bifurcations in the system, we rewrite it as a system of first order equations. Since they are nonlinear, we will linearize the system for arbitrary coordinates $x, y = \dot{x}$ and use the linearized system to assess the stability of the fixed points. By searching for locations in (μ, a) parameter space where the stability (more explicitly, the eigenvalues / discriminant) changes abruptly, we can locate the Hopf bifurcation as a function of μ and a.

We begin by rewriting

$$\ddot{x} + \mu(x^2-1)\dot{x} + x - a = 0 \qquad ,$$

as

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= \mu y - \mu x^2 y - x + a \end{aligned}$$

by letting $y = \dot{x}$. Thus, the linearized system is described by the matrix

$$\mathbb{A}(x, y) = \begin{pmatrix} 0 & 1 \\ -2\mu xy - 1 & \mu(1 - x^2) \end{pmatrix} \quad .$$

Now, we search for fixed points of the system. By inspection, it is obvious that $y^* = 0$ only due to the equation for $\dot{x}$. This leaves only $x^* = a$ as the value for the fixed point in the x coordinate. Thus, we have a lone fixed point at $(x^*, y^*) = (a, 0)$. Evaluating our linearized system at this location in the $x - y$ plane yields

$$\mathbb{A}(a, 0) = \begin{pmatrix} 0 & 1 \\ -1 & \mu(1 - a^2) \end{pmatrix} \quad .$$

This implies that

$$\begin{aligned} \operatorname{Tr} \mathbb{A}(a, 0) &= \mu(1 - a^2) \\ \det \mathbb{A}(a, 0) &= 1 \end{aligned}$$

The positive-definite determinant for the system at the fixed point implies *either* real eigenvalues with the same sign (node) or a complex conjugate pair. To determine this, we compute the eigenvalues

$$\lambda_{\pm} = \frac{\mu(1 - a^2)}{2} \pm \frac{1}{2}\sqrt{\mu^2(1 - a^2)^2 - 4} \qquad .$$

From this, we can readily calculate the values in parameter space where the discriminant vanishes and the eigenvalues change domain. This occurs when

$$\mu^2(1 - a^2)^2 - 4 = 0 \qquad .$$

From this, we have through routine manipulation that

$$\begin{aligned} \mu^2(1 - a^2)^2 - 4 &= 0 \\ \implies \sqrt{\mu^2(1 - a^2)^2} &= 2 \\ \implies \mu(1 - a^2) &= 2 \\ \implies \mu &= \frac{2}{1 - a^2} \\ \text{or} \qquad a &= \pm\sqrt{1 - \frac{2}{\mu}}, \end{aligned}$$

defining the curves in (μ, a) space that trigger a Hopf bifurcation in the system.

4 Problem 8.2.9: Predator-Prey Model

Consider the predator-prey model

$$\dot{x} = x\left(b - x - \frac{y}{1 + x}\right) \quad , \qquad \dot{y} = y\left(\frac{x}{1 + x} - ay\right) \qquad ,$$

where $x, y \geq 0$ are the population values and $a, b \in \mathbb{R}^+$ are parameters.

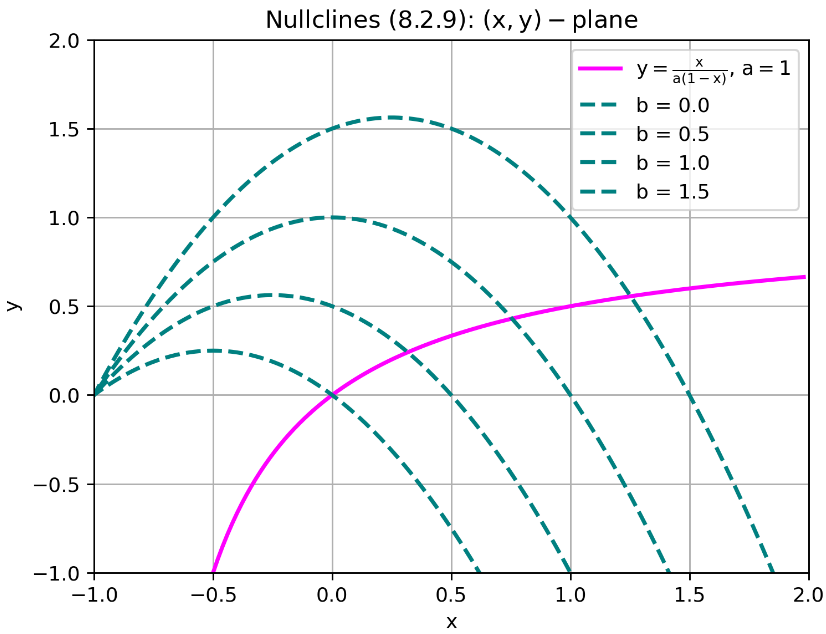

Figure 2: Nullclines of the predator-prey model from Section 4. The nullclines for $\dot{x} = 0$ are shown in teal, labeled according to the varying parameter b. The nullcline representing $\dot{y} = 0$ (magenta) is fixed for a single value of $a = 1$.

4.1 (a) Sketch the nullclines and discuss the bifurcations that occur as b varies.

The nullclines of the system are shown in Figure 2. The nullcline for $\dot{x} = 0$ is shown in teal for varying values of $b \in [0, 1/2, 1, 3/2]$. As seen in the Figure, with increasing b the intersection of the two nullclines occurs at greater values of x. The nullcline for $\dot{y} = 0$ is shown in magenta, and a value of $a = 1$ was assumed. For larger values of a, the horizontal slope of the nullcline is "flattened". For values of a smaller than unity, the slope of the nullcline is increased until it approaches vertical (undefined) at $a = 0$. Hence, regardless of the values of a, b, there is a single intersection of the nullclines and the system admits a single fixed point. As b is increased from $b = 0$, the fixed point emanates from the origin $(x^*, y^*) = (0, 0)$ to a value in the positive quadrant, representing the bifurcation to the system behavior.

Note that since $a, b > 0$ and $x, y > 0$, we have restricted all of the above discussion and all of the following analysis to the positive quadrant.

4.2 (b) Show that a positive fixed point exists for all $a, b > 0$. Don't try to find the fixed point explicitly; use a graphical argument instead.

As discussed in section 4.1 (see above), there is only one possible intersection of the nullclines in the positive quadrant. Hence, regardless of the values of a, b, the system admits one fixed point. As b is increased , the fixed point emanates from the origin $(x^*, y^*) = (0, 0)$ to a value in the positive quadrant. Note that since $a, b > 0$ and $x, y > 0$, we have restricted all of the above discussion and all of the following analysis to the positive quadrant. This can be verified by inspection of Figure 2.

4.3 (c) Show that a Hopf bifurcation occurs at the positive fixed point if,

$$a = a_c = \frac{4(b-2)}{b^2(b+2)} \qquad ,$$

and $b > 2$.

The matrix of stability for this system reads

$$\mathbb{A}(x,y) = \begin{pmatrix} b - 2x - \frac{y}{x+1} + \frac{xy}{(x+1)^2} & -\frac{x}{1+x} \\ \frac{y}{(x+1)^2} & \frac{x}{1+x} - 2a \end{pmatrix} \qquad .$$

This system then admits the trace

$$\operatorname{Tr}\mathbb{A} = b - 2x - \frac{y}{x+1} + \frac{xy}{(x+1)^2} + \frac{x}{1+x} - 2a \qquad .$$

According to Strogatz, a necessary condition for a Hopf bifurcation to occur in a system is for the trace to vanish, i.e., $\operatorname{Tr}\mathbb{A} = 0$. Then, we seek x^*, y^* such that

$$b - 2x - \frac{y}{x+1} + \frac{xy}{(x+1)^2} + \frac{x}{1+x} - 2a = 0 \qquad .$$

After some manipulation, this becomes

$$a(b - 2a - 2x)(x+1)^3 + ax(x+1)^2 - x = 0 \qquad ,$$

using the fact that y^* must satisfy

$$y = \frac{x}{a(1+x)} \qquad .$$

After further manipulation and via Strogatz, we find that

$$x^* = \frac{b-2}{2} \qquad ,$$

in order to enforce that the trace of the system passes through zero. As this occurs, the stability of the spiral changes from stable to unstable, and a limit cycle emanates outward from the fixed point at a radius corresponding to the magnitude of the real part of the eigenvalues, i.e., the absolute value of the trace. Therefore, it grows to larger radii as the parameter continues to vary.

We find from the original system equations that at the fixed point,

$$y^* = \frac{x^*}{a(1+x^*)} \qquad ,$$

which implies that

$$b - x^* - \frac{x^*}{a(1+x^*)^2} = 0 \qquad .$$

Then,

$$a(b - x^*)(1+x^*)^2 - x^* = 0 \qquad .$$

Substituting our value of $x^*(b)$, we find

$$a(b/2 + 1)(b/2)^2 = b/2 - 1$$

which eventually yields an expression for a_c:

$$a_c = \frac{b-2}{(b+2)\left(\frac{b}{2}\right)^2} = \frac{4(b-2)}{b^2(b+2)}$$

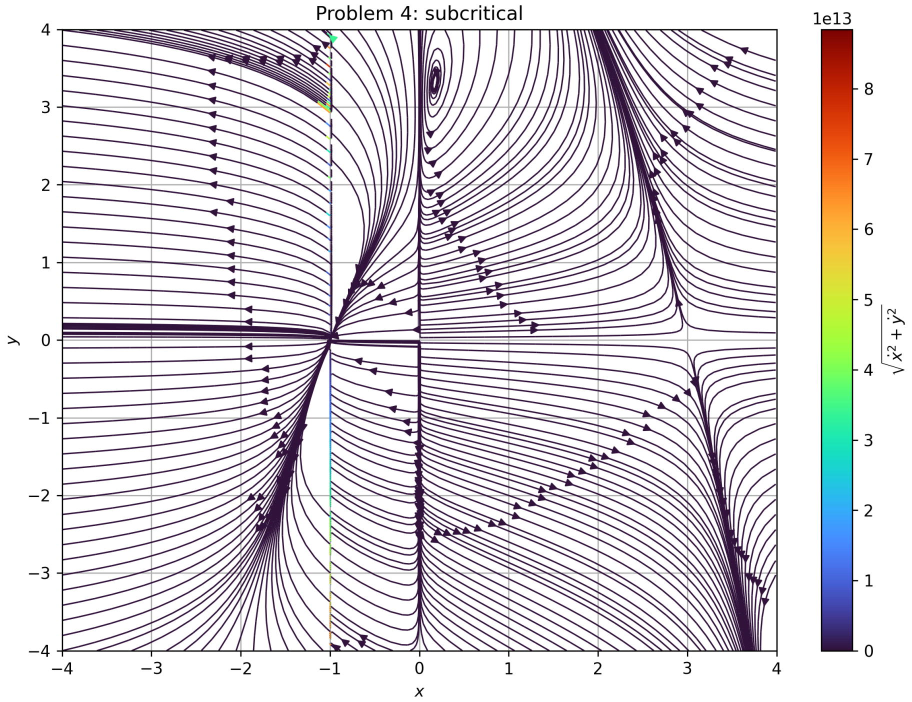

Figure 3: Subcritical case

4.4 (d) Using a computer, check the validity of the expression in (c) and determine whether the bifurcation is subcritical or supercritical. Plot typical phase portraits above and below the Hopf bifurcation.

5 Problem 8.3.1: Brusselator

The Brusselator is a simple model of a hypothetical chemical oscillator, named after the home of the scientists who proposed it. This is a common joke played by the chemical oscillator community; there is also the "Oregonator," "Palo Altonator," etc. In dimensionless form, its kinetics are

$$\dot{x} = 1 - (b+1)x + ax^2y \quad , \qquad \dot{y} = bx - ax^2y \qquad ,$$

where $x, y \geq 0$ are the dimensionless concentrations and $a, b \in \mathbb{R}^+$ are parameters.

5.1 (a) Find and classify all the fixed points.

To do so, we write the matrix of stability for the system:

$$\mathbb{A}(x,y) = \begin{pmatrix} 2axy - b - 1 & ax^2 \\ b - 2axy & -ax^2 \end{pmatrix} \quad .$$

To calculate the roots of the system, we first write the nullclines:

$$0 = 1 - (b+1)x + ax^2y \qquad ; \qquad 0 = bx - ax^2y \tag{4}$$

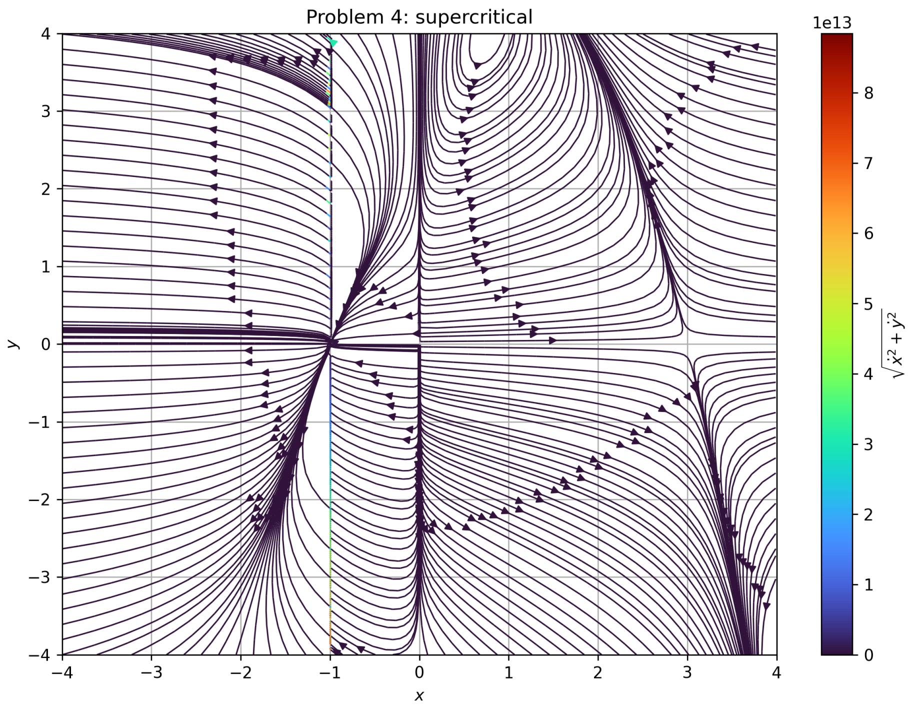

Figure 4: Supercritical case

after some manipulation of the left equation, we find that

$$y = \frac{(b+1)x - 1}{ax^2} \qquad .$$

Then, we turn to the second (right) expression in equation (1). This yields

$$y = \frac{b}{ax} \qquad ,$$

and we have expressed each of the nullclines as a function $y(x)$. Then, we can readily plot them in section 5.2. However, now we must combine these expressions to determine a value of x^* representing the fixed point. We can then use either expression from equation (4) to determine y^*, after which we will know all fixed points.

Doing so yields

$$\frac{b}{ax} = \frac{(b+1)x - 1}{ax^2} \qquad ,$$

implying that

$$bx = (b+1)x - 1 \qquad .$$

From here we find $x^* - 1 = 0 \implies x^* = 1$. Via equation (4), we have also determined that $y^* = b/a$.

Thus, we evaluate the matrix of stability $\mathbb{A}(x^*, y^*)$:

$$\mathbb{A}(1, \frac{b}{a}) = \begin{pmatrix} b-1 & a \\ -b & -a \end{pmatrix} \qquad .$$

Then, we can find

$$\mathrm{Tr}\,\mathbb{A}(1, \frac{b}{a}) = b - a - 1 \qquad ; \qquad \det \mathbb{A}(1, \frac{b}{a}) = a \qquad .$$

Therefore, we can characterize the behavior near the fixed point. Since the determinant is always positive ($a > 0$), we know that the fixed point is either a node or spiral. From equation (1), the eigenvalues have a discriminant equal to

$$D = b^2 + a^2 + 1 - 2ab - 2b - 2a \qquad ,$$

of which is not straightforward to determine the surfaces of (a, b) that satisfy.

Thus, we take a different approach. We start by setting $b = 1$ and note that this reduces to $a^2 - 4a = 0$. This is true when $a = 0$ or when $a = 4$. Likewise, setting $a = 1$ admits only one valid solution for $b > 0$, namely $b = 4$. We see then the nature of the stability for this fixed point: for combinations of a, b that are collectively small enough for the discriminant to be negative, the system behaves like a stable spiral around the fixed point in the positive quadrant. There is one direction (horizontal, in x) that is faster when a is large. When a is very large, the trajectories move much quicker along y, and much more slowly along x. Alternatively, when b is large, the vertical direction along the y-axis is the "slow direction", and trajectories "fall" towards this attractive manifold much faster along y than they fall along x. When $a = 4, b = 1$ the stable spiral converts into a stable node. In the event that b grows beyond a certain value compared to a, the spiral becomes unstable since the trace becomes large and positive. This is the supercritical destabilization. The system metamorphasizes from a stable spiral to an unstable spiral, giving rise to a limit cycle via a supercritical Hopf bifurcation.

5.2 (b) Sketch the nullclines, and thereby construct a trapping region for the flow.

The nullclines for this system, and hence, the corresponding trapping region, are included in Figure 5.

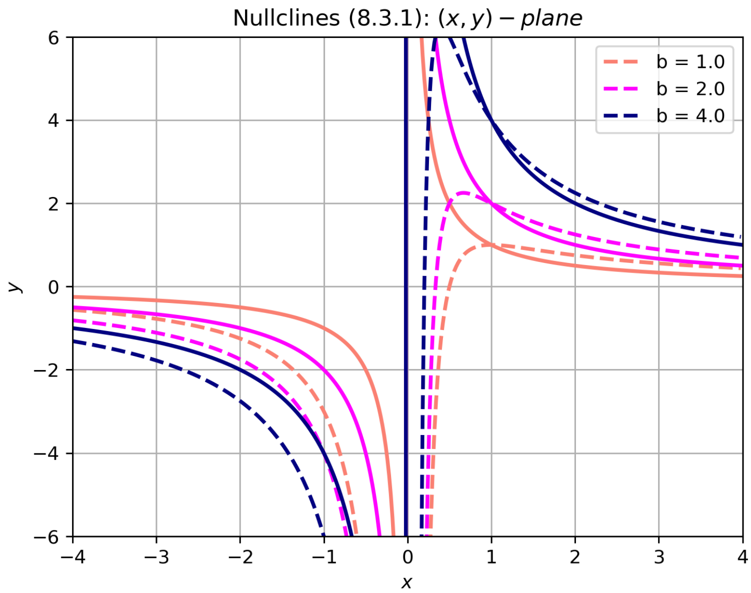

Figure 5: Nullclines for the system in Section 5. The dashed lines correspond to the curve for $\dot{x} = 0$, and the solid lines correspond to $\dot{y} = 0$. Hence, by inspection, there is a trapping region for each combination of a, b in the area bounded by the piecewise continuation of $\dot{x} = 0$ and $\dot{y} = 0$. The value of b is varied over $1, 2, 4$, and the value of a is fixed at unity.

5.3 (c) Show that a Hopf bifurcation occurs at some parameter value $b = b_c$, where b_c is to be determined.

The Hopf bifurcation involves the modification of a stable spiral into an unstable spiral (spiraling outward into a limit cycle). This occurs when the real part of the eigenvalues changes from negative to positive, i.e., it passes through zero.

Hence, it is sufficient to consider when our expression for the trace of the matrix of stability, $\mathrm{Tr}\,\mathbb{A}$ vanishes. That is,

$$\mathrm{Tr}\,\mathbb{A} = b - a - 1 = 0 \qquad \Longrightarrow \qquad b_c = 1 + a \quad .$$

Therefore, as b grows from $b < b_c = 1 + a$ to $b > b_c$, the stable spiral becomes unstable and a Hopf bifurcation transpires.

5.4 (d) Does the limit cycle exist for $b > b_c$ or $b < b_c$? Explain, using the Poincaré-Bendixson theorem.

The limit cycle exists for $b > b_c$, that is, at values of $b > a + 1$ the solutions near the fixed point are unstable but approach a stable periodic solution at some amplitude $\mu = (b - a - 1)/2$ from the fixed point. This amplitude grows as the value of b grows, and thus, the bifurcation is considered supercritical.

To exercise the Poincaré–Bendixson Theorem, we consider the limit set L of some arbitrary point in the local neighborhood of the unstable spiral fixed point at x^*, y^* from section 5.1. Since L remains bounded in the neighborhood of the fixed point for all time (via the trapping region in Figure 5) the Poincaré–Bendixson theorem mandates that L is either a limit cycle, a heteroclinic cycle, or an equilibrium. A heteroclinic cycle can be ruled out since it would require a connection to an equilibrium that is a limit point for some external coordinate to the set L. Since the stability matrix for small but nonzero values of $x - x^*, y - y^*$ does not have a pair of stable eigendirections, we can rule out the notion that L is on a separate fixed point. Thus, by essentially ruling out the other cases that could "bottle up" a trajectory in this region for all time, we can show that it is a limit cycle.

5.5 (e) Find the approximate period of the limit cycle for $b \approx b_c$.

For a frequency of oscillation ω corresponding to the complex part of the eigenvalues, the approximate period is

$$T = 2\pi/\omega \approx \frac{2\pi}{\sqrt{\det \mathbb{A}}} = \frac{2\pi}{\sqrt{a}} \qquad ,$$

in the neighborhood of the fixed point $(x^*, y^*) = (1, b/a)$. Thus, as a grows, the period of the oscillations is shortened and the "turning" speeds up. On the other hand, it is not sensitive to the specific value of b, just that $b > b_c$ to trigger the unstable spiral in the first place.

6 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the plotting: this was completed using python (`numpy` and `matplotlib`).